communications earth & environment
A Nature Portfolio journal

Article

https://doi.org/10.1038/s43247-026-03602-z

The Chicxulub impact signature in East Asia

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The Cretaceous-Paleogene boundary is defined by an asteroid impact at 66 million years ago at Chicxulub, Mexico. Despite extensive documentation worldwide, this boundary has not been clearly identified in East Asia, leaving a geographic gap in evaluating the global extent of impact records. Here we report a pronounced increase in osmium concentrations accompanied by a sharp decrease in $^{187}Os/^{188}Os$ ratios at the Kawaruppu section in the Nemuro Group, Hokkaido, Japan, which are indicative of the Chicxulub impact. These geochemical signatures are consistent with the zircon U–Pb age from an immediately overlying tuff layer, and supported by magnetostratigraphic and micropaleontological data. Iridium enrichment—a diagnostic impact marker— is subdued, suggesting partial stratigraphic gap, and the missing interval is estimated by box models to be approximately 30 thousand years. Nevertheless, our results identify a key stratigraphic horizon that enables paleoenvironmental correlation across the Cretaceous-Paleogene boundary in East Asia.

A large asteroid impact at Chicxulub, Mexico, is widely regarded as the primary cause of mass extinction at the Cretaceous-Paleogene (K-Pg) boundary at approximately 66 Ma[1] (Fig. 1A). A high abundance of platinum group elements (PGEs; Os, Ir, Ru, Rh, Pt and Pd) has been documented globally in K-Pg boundary layers, providing compelling evidence for this impact event[1,2]. In particular, because Os, Ir and Ru behave as compatible elements during planetary differentiation, they are strongly depleted in the Earth's surface and show a marked contrast with undifferentiated meteorites, making them highly sensitive tracers of extraterrestrial impacts. Osmium isotope ratio ($^{187}Os/^{188}Os$) of marine sediments is also a diagnostic tracer for identifying extraterrestrial impact horizons[3]. Because the oceanic residence time of Os (10–50 kyr[4,5]) is longer than the timescale of modern ocean circulation, $^{187}Os/^{188}Os$ values are largely homogeneous throughout the global ocean, allowing correlation among geographically distant sections. Seawater $^{187}Os/^{188}Os$ ratio reflects the balance between inputs from the upper continental crust (UCC), which has more radiogenic compositions ($^{187}Os/^{188}Os \approx 1.0$–$1.5$), and mantle and extraterrestrial sources, which have less radiogenic compositions ($^{187}Os/^{188}Os \approx 0.12$–$0.13$)[6]. Across the K-Pg boundary, marine Os isotopic records display a characteristic stepwise pattern consisting of (1) a gradual decline in $^{187}Os/^{188}Os$ ratios from ~0.6 to ~0.4 beginning near the C29R/C30N magnetic reversal (~300 kyr prior to the K-Pg boundary), (2) a sharp drop to <0.2 at the K-Pg boundary (65.845 Ma[7]), and (3) a rapid recovery to ~0.4 extending into the Danian[8–10]. The initial gradual decline is commonly attributed to emplacement of Deccan Traps[8,9,11], whereas the subsequent sharp decrease reflects a massive input of extraterrestrial Os associated with the Chicxulub impact[8–10]. Although sedimentary archives spanning the K-Pg boundary have been documented worldwide, well-constrained K-Pg boundary sections have not yet been identified in the northwestern Pacific and East Asia[12], leaving a geographic gap in evaluating the global extent and preservation of impact signals.

Recently, a stratigraphic section of the Nemuro Group was discovered in the Shiranuka Hills[13], in eastern Hokkaido, Japan (Fig. 1B, C), which potentially preserves a continuous sedimentary record across the K-Pg boundary. The Nemuro Group is located in a distal site from the Chicxulub impact site (Fig. 1A) and is therefore expected to be less affected by impact-induced sedimentary disturbances, such as mass wasting or tsunami-related reworking[14]. Moreover, its high sedimentation rate near the continental margin[15] allows reconstruction of paleoenvironmental changes at high temporal resolution. Despite these advantages, no geochemical evidence for the Chicxulub impact has previously been reported in this region.

The primary aim of this study is to test whether evidence of the Chicxulub impact is preserved in East Asia. To address this question, we present high-resolution data on PGE concentrations and Os isotopic ratios to identify and constrain the stratigraphic position of the K-Pg boundary in two sections (the Kawaruppu and Mokawaruppu sections) of the Nemuro Group (Fig. 1B, C, and Supplementary Fig. 1). These geochemical data are integrated with data on paleomagnetism, microfossil assemblages and U–Pb dating of zircon to refine the stratigraphic position of the K-Pg boundary in

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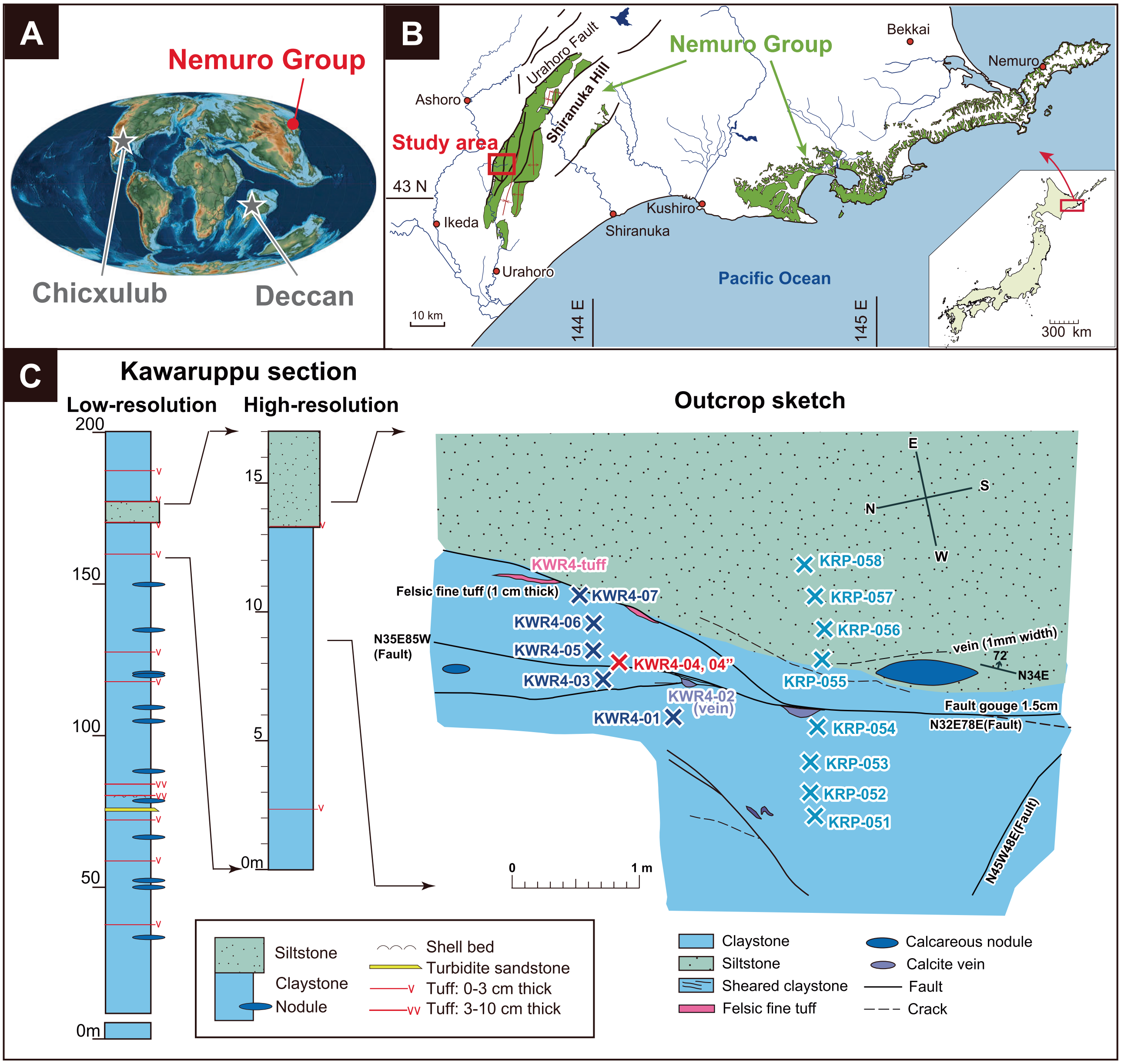

Fig. 1 | Location map and stratigraphy of the Kawaruppu section. **A** Paleogeography at 66 Ma[49] showing the location of the Nemuro Group. **B** Map showing the modern distribution of the Nemuro Group and the location of the study area. Modified from ref. 50. **C** Lithologic columns of Kawaruppu section (left) and sketch of the outcrop around the K-Pg boundary (right). Photo of this outcrop is in Supplementary Fig. 3.

this region. The Upper Cretaceous to lower Paleogene Nemuro Group is distributed along the Nemuro coast and the Shiranuka Hills, Hokkaido, Japan (Fig. 1B).

The Nemuro Group is mainly composed of hemipelagic mudstones and gravity flow deposits[16]. The main survey area for this study is a stream branching off from a tributary of the Kawaruppu River in the Shiranuka Hills (Kawaruppu section, 43.098°N, 143.730°E). This section primarily consists of homogenous dark clay stone, sandy siltstone (171–178 m), thin silicic tuff layers and frequent calcareous nodules (Fig. 1C). The strike (N10–20°E) and dip (80–90°SE) remain consistent throughout the section, with no evidence of fracture zones. Paleomagnetic measurements at a total of 18 stratigraphic horizons throughout the section consistently show reverse polarity[17] (Fig. 1C). We collected two samples from volcanic tuff beds for U–Pb dating of zircon and 22 mudstone samples from a 200-meter-thick stratigraphic section for developing low-resolution profiles of PGE concentrations and Os isotope ratios across the entire section (the low-resolution interval shown in Fig. 1C, and Supplementary Fig. 2). For high-resolution analysis around the K-Pg boundary, a total of 38 samples (KRP) and 8 additional samples (KWR4) were collected from an approximately 10-m interval within the same section (the high-resolution interval shown in Fig. 1C, and Supplementary Fig. 2). Sedimentary rock samples were also collected from the Mokawaruppu section (43.070°N, 143.703°E; Supplementary Fig. 4), where a clay layer previously interpreted as the K-Pg boundary (Supplementary Fig. 5) was identified on the basis of biostratigraphic records[18].

Results & discussion

U–Pb geochronology

We have obtained zircon U–Pb ages of 65.781 ± 0.018 Ma for a tuff layer at 178 m, and 65.44 ± 0.89 Ma for another tuff layer at 188 m (Fig. 1C, and Supplementary Data 1 and 2). With respect to the reversed magnetic polarity of the entire section, these dates indicate that the section resides in the interval of Chron C29R (66.106–65.521 Ma; the recalculated age values reported by ref. 19, using the Fish Canyon sanidine (FCs) age from ref. 20

and the ^{40}K decay constants from ref. 21). Extrapolation from the precise age of the lower tuff leads us to expect PGE evidence for the K-Pg boundary (65.845 Ma[7]) within proximal underlying strata.

PGE and Re–Os records of the Kawaruppu section

The low-resolution record was divided into three intervals: Interval L1 (below 173.1 m), level 173.1 m (Sample KW 289), and Interval L2 (above 173.1 m) (Fig. 2A). Throughout Interval L1, Os and Re concentrations remain relatively constant at ~60–90 pg g^{-1} and ~600–800 pg g^{-1}, respectively (Supplementary Data 3). Initial Os isotope ratio ($^{187}Os/^{188}Os_i$) also shows stable value of ~0.6 in this interval. Sample KW 289 at 173.1 m has the highest Os concentrations (88, 94 and 126 pg g^{-1}, $n = 3$), and has the lowest $^{187}Os/^{188}Os_i$ values (0.29, 0.40 and 0.43, $n = 3$) in this area. In Interval L2, Os concentration decreases to ~70 pg g^{-1} (Supplementary Data 3). The $^{187}Os/^{188}Os_i$ values take stable values of ~0.4 throughout this interval.

The high-resolution record was divided into three intervals: Interval H1 (below 12.40 m), level 12.40 m, and Interval H2 (above 12.40 m) (Fig. 2B). In Interval H1, Os concentrations range from ~67 to 110 pg g^{-1} (Supplementary Data 4). From 6.58 m to 11.50 m, $^{187}Os/^{188}Os_i$ values are stable at ~0.6. At 12.0 m, the ratio suddenly drops to ~0.5 and remains stable until 12.35 m (within Interval H1). Iridium concentration continues to have similar values throughout this interval (~10–20 pg g^{-1}). We measured two samples from the same stratigraphic level at 12.40 m, KWR4-04 ($n = 3$), and KWR4-04" ($n = 2$). The maximum Os concentration and minimum $^{187}Os/^{188}Os_i$ values occur at this interval (~248 pg g^{-1} and ~0.235–0.325, respectively). Iridium concentration slightly increases to ~34 and 37 pg g^{-1}. The concentrations of Ru, Pd, Pt at 12.40 m are ~30–33, ~1612–1939 and ~589–911 pg g^{-1} respectively, which are comparable to those in the adjacent intervals (Supplementary Data 4). In Interval H2, Os concentration decreases and remains at around 90 pg g^{-1} and $^{187}Os/^{188}Os_i$ values returned to more radiogenic values of ~0.4. Iridium concentrations return to the background level of ~13–20 pg g^{-1} in this interval.

PGE and Re–Os records of the Mokawaruppu section

The clay layer at Mokawaruppu (MK), previously interpreted as the K-Pg boundary clay layer[18], was measured twice for Os (124.8–127.8 pg g^{-1}) and Re concentrations (2307 pg g^{-1}), and $^{187}Os/^{188}Os_i$ values (0.438–0.443), which are close to those values in Interval L2 and H2 in the Kawaruppu section (Supplementary Data 5). The Ir, Ru, Pd, Pt concentrations are 20, 31, 1840, 644 pg g^{-1}, respectively (Supplementary Data 5).

Geochemical evidence of the K-Pg impact in the Kawaruppu section

Our results demonstrate that the osmium isotopic ratio in the Kawaruppu section is broadly consistent with the global seawater record across the K-Pg boundary (Fig. 3). The $^{187}Os/^{188}Os_i$ values are ~0.6 in intervals L1 and H1, and ~0.4 in intervals L2 and H2 (Fig. 2A, B), corresponding to typical Maastrichtian[8,9] and Danian[8,22] seawater values, respectively. However, the timing of the decline in $^{187}Os/^{188}Os_i$ values from 0.6 to 0.4 in the latest Maastrichtian–commonly attributed to the weathering of juvenile basalt from the Deccan Traps– does not coincide with the global trend[8,9,11]. This discrepancy suggests local heterogeneity in seawater $^{187}Os/^{188}Os_i$, likely reflecting a substantial regional input of radiogenic Os derived from continental sources.

Sharp increases in Os concentrations and decreases in $^{187}Os/^{188}Os_i$ values are observed at both levels, KW 289 and KWR4-04 (Fig. 2A, B). Although KWR4-04 exhibits slightly higher Os concentrations (248 pg g^{-1}) and lower $^{187}Os/^{188}Os$ values (0.235–0.325) than KW289, these two horizons can be correlated macroscopically in the outcrop. We therefore interpret them as stratigraphically equivalent, or at least as representing very closely spaced horizons. Such high Os concentrations coupled with low $^{187}Os/^{188}Os_i$ values are characteristic geochemical features of the K-Pg boundary layer[8,9], indicating that the stratigraphic level of 12.40 m (KWR4-04) was deposited at or near the time of the K-Pg impact. The Os concentrations and $^{187}Os/^{188}Os_i$ values in the KRP and KWR4 samples plot along a mixing curve between CI chondrite and the UCC compositions (Fig. 4A). This trend suggests that the observed variation can be explained by an increased relative contribution of Os from an extraterrestrial source. The extremely low PGE concentrations measured in the tuff layer (KWR4-tuff) and carbonate vein (KWR4-02) indicate that local volcanic or hydrothermal inputs exerted a negligible influence on PGE variations in this section (Fig. 2B).

An independent chronological constraint is provided by U-Pb dating of zircon of a tuff layer located approximately 5 m above level KWR4-04 (Fig. 2A). Our interpretation that KWR4-04 was deposited at or near the time of the K-Pg impact (65.845 ± 0.053 Ma) is consistent with the U–Pb age obtained for this tuff (65.781 ± 0.018 Ma). A detailed discussion of sedimentation rates is provided in Supplementary Note 1 and Supplementary Fig. 6.

Slightly elevated Ir concentrations (~17–37 pg g^{-1}) are also observed at level KWR4-04, coincident with the Os maximum. However, this Ir enrichment is modest compared to that reported from other K-Pg boundary sections, such as Caravaca in Spain or Stevns Klint in Denmark, where Ir concentrations reach on the order of tens of ng g^{-1}[23]. Concentrations of Ru and Pd show no increase at KWR4-04 (Supplementary Data 4), while Pt concentrations are only slightly elevated relative to background levels and show comparable increases at other stratigraphic levels (Supplementary Data 4). Furthermore, all KWR4-series samples display UCC-like CI-chondrite-normalized patterns for Ir, Ru, Pt and Pd (Fig. 4B), indicating that the extraterrestrial contribution for those elements was limited relative to that of Os.

Why do only the Os concentrations and $^{187}Os/^{188}Os_i$ ratios in KWR4-04 indicate an extraterrestrial contribution? Three possible explanations can be considered: (1) redistribution by physical mixing, (2) diagenetic remobilization, and (3) partial loss of sediment. Redistribution of Ir[24] and Os[10] has been reported at K-Pg boundary sections in various regions. In many cases, this redistribution has been attributed to physical mixing mainly caused by bioturbation (1). However, such physical mixing would redistribute all PGEs simultaneously and would not produce selective remobilization. Therefore, it is inconsistent with the selective Os enrichment observed in KWR4-04. Diagenetic remobilization (2) typically involves the selective dissolution or precipitation of redox-sensitive elements in response to changing depositional redox conditions. However, the seven samples from the Kawaruppu section, including KWR4-04, show nearly identical lithology and major element composition (Supplementary Fig. 7 and Supplementary Data 6), suggesting stable depositional conditions and providing little evidence for redox-driven diagenetic modification.

We therefore infer that partial sediment loss (3), namely a stratigraphic gap, is the most plausible explanation. In this scenario, the PGE abundances can be explained by differences in recovery patterns of individual PGEs following the K-Pg impact, governed by their respective residence times in the ocean. Estimated residence times of PGEs are: 10–50 kyr[4,5] for Os, 2–20 kyr[25] for Ir, 10–30 kyr[26] for Ru, 24 ± 10 kyr[26] for Pt, and ~30[26] for Pd. Following the Chicxulub impact event, Os, which has a relatively long oceanic residence time, would retain an impact-derived concentration anomaly and isotopic signature in seawater for a longer duration, whereas other PGEs would have been removed from the ocean more rapidly. We interpret KWR4-04 as having been deposited during this interval. Sediments deposited at and immediately after the Chicxulub impact may not be preserved in the Kawaruppu section due to a hiatus or removal by faulting. We infer that this stratigraphic interval has been partially lost due to a small fault running near sample KWR4-04 (Fig. 1C).

Estimation of the missing interval at the K-Pg boundary

As discussed above, our geochemical data strongly suggest that the Kawaruppu section preserves a sedimentary record spanning the K-Pg boundary, although the sedimentary horizon corresponding to the exact timing of the Chicxulub impact is missing due to a short stratigraphic gap. We estimate the duration of the missing interval based on Os/Ir weight ratios in the sediments. Because chondritic materials have a much lower Os/Ir ratio (~1.0[27]) than that of modern seawater (~100[23]) and typical marine

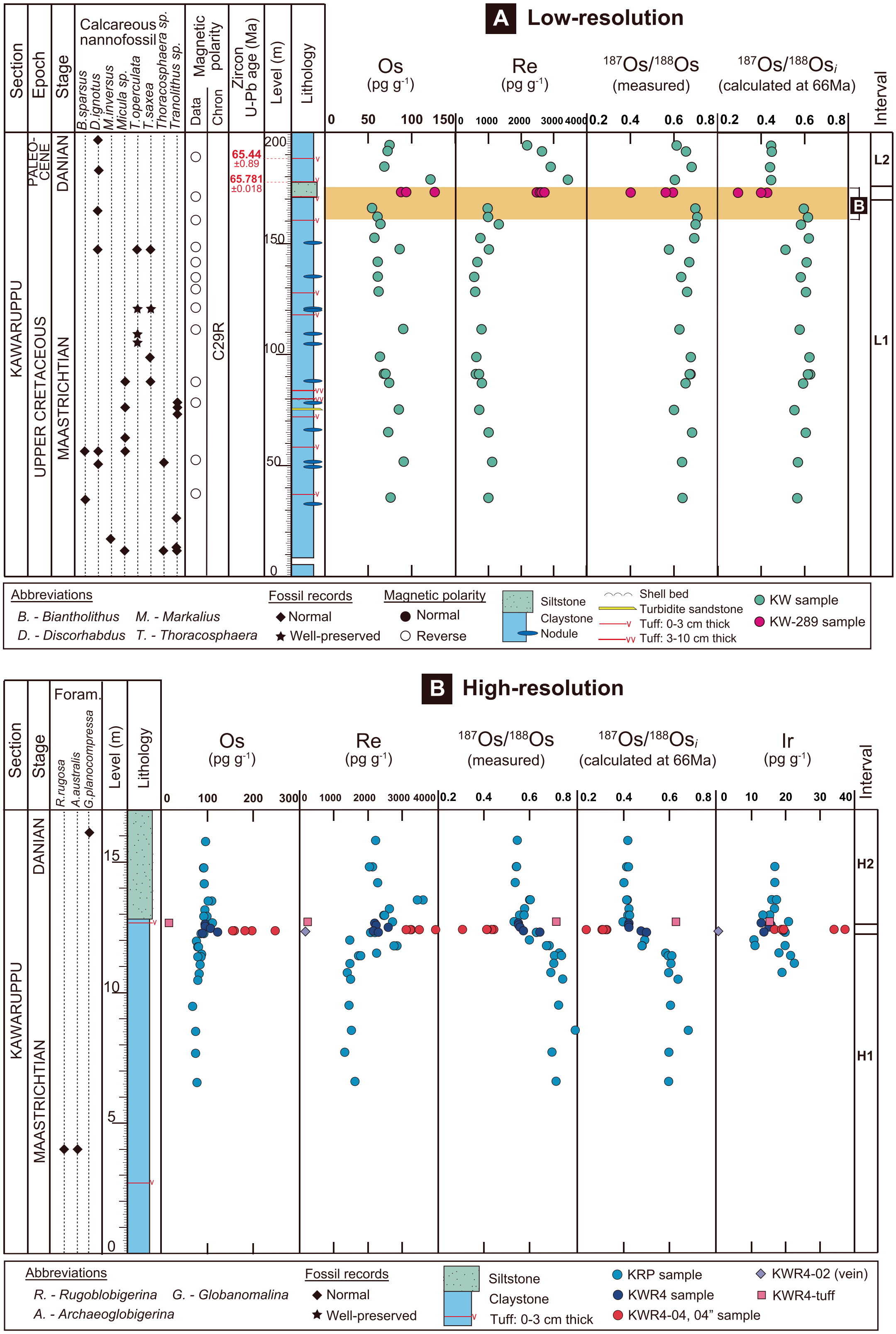

Fig. 2 | Stratigraphic profiles of $^{187}Os/^{188}Os$ ratio and PGE concentrations of Kawaruppu section. **A** Low-resolution records of Re and Os concentrations and $^{187}Os/^{188}Os$ ratio. **B** High-resolution records of Re, Os and Ir concentrations and $^{187}Os/^{188}Os$ ratio. The orange horizontal bar in **A** corresponds to the range of **B**. Lithology and fossil records are after ref. 13. Magnetic polarity data is after ref. 17. Details of the PGE concentrations and $^{187}Os/^{188}Os_i$ values are in the Supplementary Data 3 and 4.

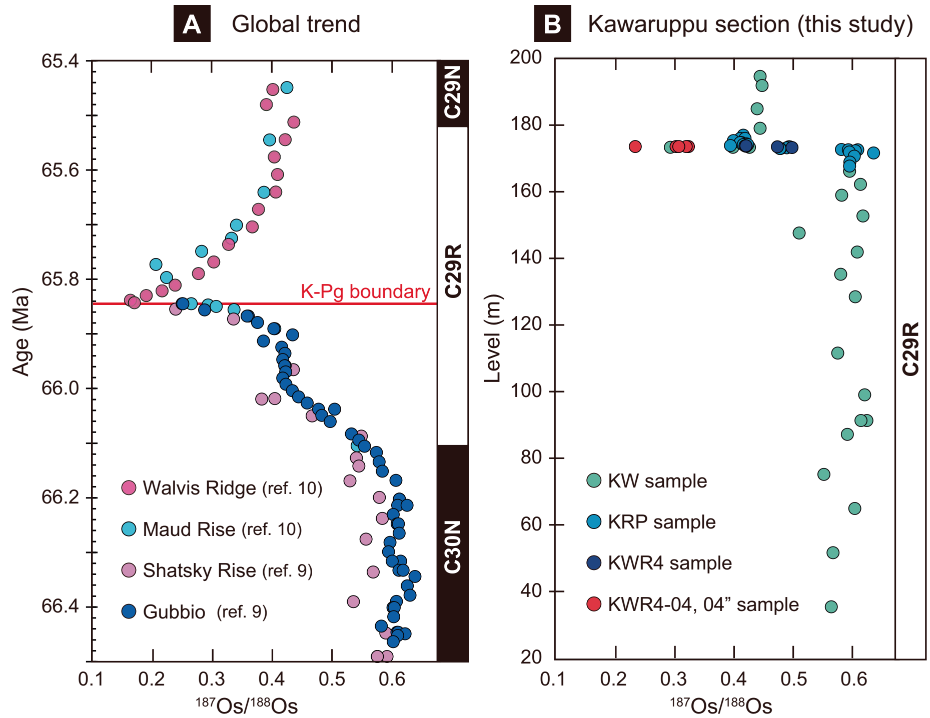

Fig. 3 | Global compilation of marine Os isotopic records and comparison with the Kawaruppu section. A $^{187}Os/^{188}Os$ ratios in distal sites across the K-Pg boundary[9,10]. The red horizontal line represents the biotsratigraphic K-Pg boundary. All Os isotopic data are integrated using the following age constraints: the Chron C29r (66.106–65.521 Ma) and the K-Pg boundary (65.845 Ma) (see the U–Pb geochronology section for details). **B** $^{187}Os/^{188}Os$ ratios in the Kawaruppu section. The "low-resolution" and "high-resolution" datasets were integrated by correlating the horizons KW289 and KWR4-04, which show a pronounced decrease in the $^{187}Os/^{188}Os$ ratio.

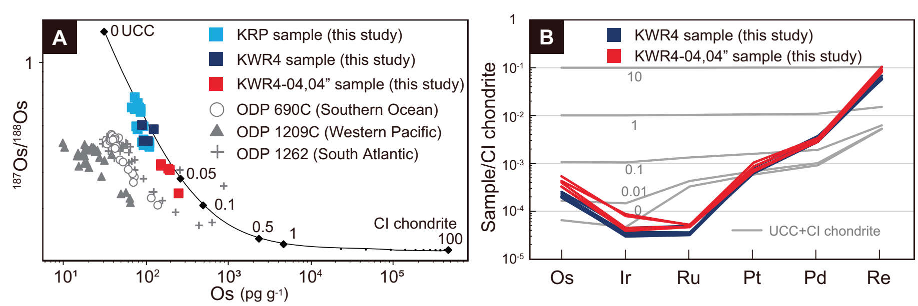

Fig. 4 | PGE compositions of KRP samples. A $^{187}Os/^{188}Os_i$ values versus Os concentrations (logarithmic scale) from this study (high-resolution series) and previously published results[10] (gray scale). The curve represents the calculated mixing line between the upper continental crust (UCC) and CI chondrite. UCC[51]: Os = 31 pg g^{-1} and $^{187}Os/^{188}Os$ = 1.4. CI chondrite[48]: Os = 460.5×10^3 pg g^{-1} and $^{187}Os/^{188}Os$ = 0.126. The numbers on the curve show percentages of admixture of CI chondrite (wt%). **B** CI chondrite-normalized PGE patterns of KWR4 samples. The UCC and CI chondrite values are the same reference values as **A**. The numbers on the gray lines show percentages of admixture of CI chondrite (wt%).

sediments (~2–7[28,29]), a large asteroid impact such as the Chicxulub event causes an abrupt decrease in Os/Ir ratio of sediments[3,23]. Immediately after the impact, a transient increase in the Os/Ir ratios is expected[3] before a return to pre-impact values, owing to the more rapid removal of Ir from the water column relative to Os. Such transient increases have indeed been documented in marine sediments from the Gulf of Mexico (Site M0077, Guayal, El Mulato and La Lajilla sections)[30], where Os concentration decreased more slowly than those of Ir and Ru following the Chicxulub impact (Supplementary Fig. 8), resulting in temporarily elevated Os/Ir ratios (Supplementary Fig. 9). Importantly, their data suggest that the residence times of Os and Ir are precisely estimated to be 40 kyr and 6 kyr, respectively (Supplementary Fig. 10). Consistent with these empirical observations, numerical simulations by Paquey et al.[3] demonstrated that sedimentary Os/Ir ratios exhibit a transient increase following the late Eocene impact.

In the Kawaruppu section, no stratigraphic horizon exhibits the combination of unradiogenic $^{187}Os/^{188}Os$ values and low Os/Ir ratios expected for sediments deposited at the time of the Chicxulub impact itself (Fig. 5A1, 5A2). Instead, markedly elevated Os/Ir ratios (~10) at KWR4-04 (Fig. 5A2) indicate that this horizon was deposited during the transient increase phase shortly after the Chicxulub impact. To estimate the timing of this Os/Ir transient phase, we developed a simple one-box model (hereafter referred to as Model A; see Method for details) (Fig. 5B1, 5B2). In Model A, the duration and amplitude of the transient Os/Ir increase are primarily controlled by (i) Os and Ir concentrations in sediments at the time of Chicxulub impact (t = 0), (ii) their post-impact steady-state concentrations constrained by measured sediment data, (iii) the sedimentation rate and (iv) the residence times of Os and Ir in the ocean. When we applied the following plausible parameter ranges–(i) Os = 284–426 pg g^{-1}, Ir = 414–621 pg g^{-1}, (ii) Os = 98 pg g^{-1}, Ir = 16 pg g^{-1}, (iii) sedimentation rates of 60–90 cm kyr^{-1} and (iv) residence times of 40 kyr for Os, 6 kyr for Ir–the model reproduced a transient increase in sedimentary Os/Ir ratios to values > 10 approximately 20–30 kyr after the Chicxulub impact (Fig. 5B2).

By using the parameters constrained by Model A, we developed a complementary box model to reproduce the observed variations in all analyzed PGE concentrations (Os, Ir, Ru, Pt and Pd) following the Chicxulub impact (hereafter referred to as Model B; see Supplementary Note 2 for details). Model B yields element-specific decreases in PGE concentrations governed by their respective oceanic residence time (Supplementary Fig. 11). When parameters were fixed at (i) Os = 370 pg g^{-1} and Ir = 538 pg g^{-1} at t = 0, and (iii) sedimentation rate of 69 cm kyr^{-1}, the modeled

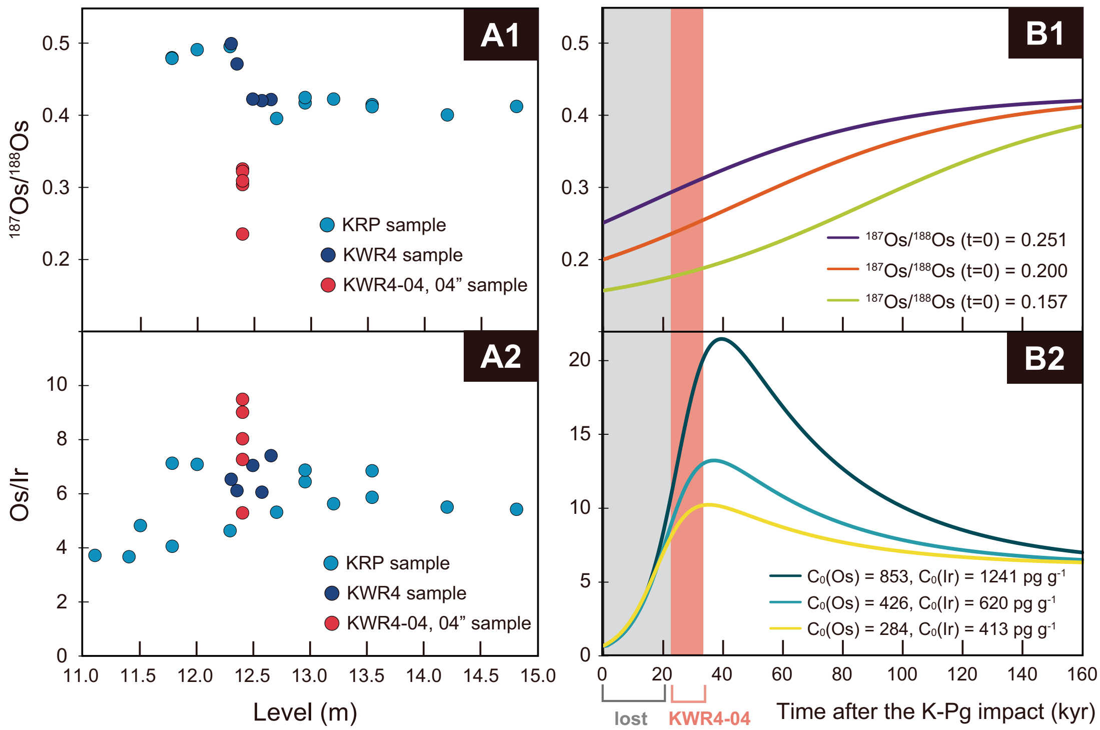

Fig. 5 | Estimated depositional timing of KWR4-04. A ^{187}Os/^{188}Os$_i$ (**A1**) and Os/Ir ratio (**A2**) of the Kawaruppu section around the K-Pg boundary. **B** Simulated temporary variations of ^{187}Os/^{188}Os (**B1**) and Os/Ir ratio (**B2**) of the sediment after the Chicxulub impact. The red and gray vertical bands indicate the inferred stratigraphic gap and the interval when KWR4-04 is suggested to have been deposited, respectively. Note that $t = 0$ represents the time of the Chicxulub impact. $C_{0(\mathrm{Os})}$ and $C_{0(\mathrm{Ir})}$ denote the Os and Ir concentrations at $t = 0$, respectively (see the "Calculation of Os/Ir and ^{187}Os/^{188}Os ratios" section in detail).

PGE concentrations at ~30 kyr after the impact closely match those observed in Sample KWR4-04. Sample KWR4-04 therefore represents sediment deposited approximately 30 kyr after the Chicxulub impact, when Ir and Ru concentrations had nearly returned to pre-impact background levels, whereas elevated Os concentrations and the associated less radiogenic ^{187}Os/^{188}Os ratio still persisted in the ocean. The results of these two models indicate that missing stratigraphic interval at the Kawaruppu section spans approximately 30 kyr at and after the K-Pg boundary.

Microscopic observation of thin sections of KWR4-04 reveal no other evidence of impact-related materials, such as shocked quartz, spherules, or impact glass tektites (Supplementary Fig. 12), which are commonly reported from well-preserved K-Pg boundary sections in other regions[31]. The lack of shocked minerals and spherules may further suggest that the sediment layer corresponding to the time of the impact was removed by subsequent fault activity.

Re-investigation of the Mokawaruppu section

The K-Pg boundary in the Mokawaruppu section has previously been interpreted as representing a complete stratigraphic succession based on paleontological evidence[18]. A thin clay layer has long been regarded as the K-Pg boundary clay layer and represents the only reported expression of the K-Pg boundary strata in East Asia[12]. However, this clay layer has not previously been analyzed for Ir and other PGE concentrations, nor ^{187}Os/^{188}Os ratios. In this study, we conducted detailed field surveys in this section and measured PGE concentrations and ^{187}Os/^{188}Os ratio of the clay layer (hereafter referred to as MK) to compare them with data from the Kawaruppu section (Supplementary Fig. 1).

The Kawaruppu and Mokawaruppu sections share similar lithofacies dominated by nearly homogeneous mudstone, reflecting deposition within the same sedimentary basin[16]. A key difference between the two sections is the presence of the MK in the Mokawaruppu section. Notably, strata immediately above and below MK are heavily brecciated, forming a fracture zone (Supplementary Fig. 5); the dip and strike of the clay layer differ markedly from those of the surrounding outcrops (Supplementary Fig. 4). These geological observations imply that MK may not represent a coherent depositional unit, but instead was likely formed by post-depositional deformation.

Geochemically, the MK sample shows neither PGE anomalies nor the decrease in the ^{187}Os/^{188}Os ratios that are characteristic of the K-Pg boundary (Supplementary Data 5). Instead, the ^{187}Os/^{188}Os ratio of MK (~0.441) closely matches that of the Danian interval in the Kawaruppu section (~0.428), and PGE concentrations are nearly identical to the background levels observed at the Kawaruppu section. Based on the stratigraphic framework and PGE records, we interpret that the outcrop previously identified as the K-Pg boundary clay layer in the Mokawaruppu section does not preserve the Chicxulub impact ejecta layer. Because geochemical analyzes in the Mokawaruppu section were limited to the MK horizon, it remains possible that the K-Pg boundary is preserved at stratigraphic levels below this horizon.

Conclusions

We investigated the stratigraphy around the K-Pg boundary in the Nemuro Group based on PGE compositions and Os isotopic ratios. Constrained by a zircon U–Pb age for an overlying volcanic tuff (65.781 ± 0.018 Ma), our results show a temporary increase in Os concentrations and a distinct decrease in the ^{187}Os/^{188}Os values in the Kawaruppu section that are attributed to the supply of less radiogenic meteoritic Os derived from the Chicxulub impact. However, the section lacks a clear peak of Ir, the conventional marker of the K-Pg impact, which suggests a short stratigraphic gap at and above the K-Pg boundary. Box model estimates based on Os/Ir ratios place this gap at approximately 30 kyr. Despite this stratigraphic gap, our findings provide definitive geochemical evidence of the Chicxulub impact in the East Asia region. Given the high sedimentation rate in this section (>30 cm kyr^{-1}), these stratigraphic data establish a robust

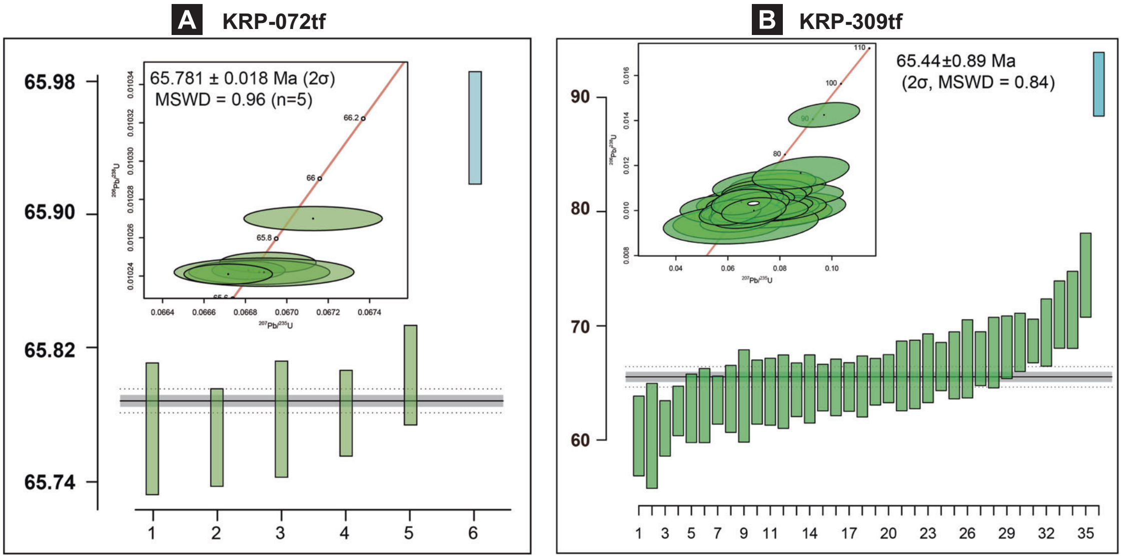

Fig. 6 | U–Pb concordia and ^{238}U–^{206}Pb age distribution for zircons, along with weighted mean ages. A KRP-072 (CA-ID-TIMS method); **B** KRP-309 (LA-ICPMS method). The weighted mean age is shown by the solid gray horizontal line, while the dashed gray lines indicate the 2σ uncertainty envelope of the weighted mean. Individual analyses are represented by green vertical bars, the heights of which correspond to their respective 2σ analytical uncertainties (error bars) on the U–Pb dates. The blue vertical bar denotes analyses that were statistically rejected and excluded from the weighted mean calculation. In the concordia diagrams, green ellipses represent individual analyses with their 2σ uncertainty ellipses.

foundation for future high-resolution analyzes of paleoclimatic, paleoceanographic, and paleoecological changes across the K-Pg boundary.

Methods

Zircon U–Pb measurements

Two tuff beds were sampled for radioisotopic dating from 178 m (KRP-072) and 188 m (KRP-309) levels, respectively. Approximately 200 g of each tuff sample was crushed using a jaw crusher and then separated into magnetic and nonmagnetic minerals with a neodymium magnet. The nonmagnetic minerals were further separated into light and heavy fractions using sodium polytungstate. Zircon grains were selected from the heavy fractions under a binocular microscope.

For KRP-072, U–Pb zircon CA-ID-TIMS dating was performed at Boise State University (BSU) Isotope Geology Laboratory. Zircon crystals were placed in muffle furnaces at 900 °C for 60 h to anneal radiation-damaged domains[32]. Individual crystals were mounted in epoxy, polished to center zones and imaged by cathodoluminescence (CL). From the compiled images, spots were selected for laser ablation–inductively coupled plasma mass spectrometry (LA-ICP-MS) using a Photon Machines Excite 193 nm Excimer laser and ThermoElectron ICAP-RQ quadrupole ICPMS. Based on CL images and LA-ICP-MS data, single zircon grains were plucked from their respective epoxy mounts for the subsequent CA-ID-TIMS analysis. Individual zircon grains were chemically abraded in concentrated (29 M) HF at 190 °C for 12 h to mitigate open-system behavior and remove inclusions[32]. The residual zircon grains were spiked with the EARTHTIME mixed ET2535 tracer solution[33,34] and completely dissolved in 29 M HF at 220 °C for 48 h in Parr bombs. Following dissolution, the zircon+tracer solutions were dried to fluorides, and then re-dissolved in 6 M HCl at 180 °C overnight. Uranium and Pb were separated by anion-exchange column chromatography using 50 µl columns and AG-1 ×8 resin[35]. The U and Pb isotopic measurements were performed on an IsotopX Phoenix X62 multi-collector thermal ionization mass spectrometer. Determined U–Pb dates and uncertainties for each analysis were calculated from isotope ratio measurements using the algorithms of ref. 36 and the U decay constants of ref. 37 (Fig. 6).

For KRP-309, U–Pb data were obtained using laser ablation (LA, NWR213 Electro Scientific Industries) coupled with an inductively coupled plasma mass spectrometer (ICP-MS Agilent 7700×) at Nagoya University, Japan. The LA-ICP-MS analysis procedures are detailed in refs. 38,39. This was analyzed using 91500[40] as the primary zircon standard and Plesovice as the secondary zircon standard. Common P correction was achieved using the Isoplot R program[41] as a concordia projection of the assumed value of an initial Pb composition from ref. 42. The weighted mean age of Plesovice ($N = 5$) yielded a ^{238}U–^{206}Pb age of 337.6 ± 6.3 Ma (95% CI; MSWD = 1.1) and ^{235}U–^{207}Pb age of 320 ± 9.4 Ma (95% CI; MSWD = 0.74), which are consistent with the weighted mean ^{238}U–^{206}Pb age (337.13 ± 0.37 Ma) of ID-TIMS data[43]. The Isoplot R program[41] was employed to calulate weighted means, mean square weighted deviations (MSWDs), and probabilities as well as to plot data on the concordia diagram (Fig. 6).

Re and PGE analysis

We measured Os concentrations and ^{187}Os/^{188}Os ratio using negative thermal ionization mass spectrometry (N-TIMS; TRITON, Thermo Fisher Scientific) at JAMSTEC and Re, Ru, Pd, Ir, Pt concentrations using an inductively coupled plasma quadrupole mass spectrometer (ICP-QMS; iCAP Q, ThermoFisher Scientific) at JAMSTEC. The methods used for sample digestion, chemical purification, and mass spectrometry are based on procedures outlined in refs. 30,44.

Approximately 0.5 g of powdered samples were weighted, and appropriate amounts of mixed PGE and Re spikes (^{185}Re, ^{190}Os, ^{99}Ru, ^{105}Pd, ^{191}Ir, ^{196}Pt) were added to a quartz tube. After adding 4 ml of inverse aqua regia (mixture of 1 ml of 30 wt% HCl: TAMAPURE-AA-100 and 3 ml of 68 wt% HNO_3: TAMAPURE-AA-10 from Tama Chemicals Co. Ltd, Japan), each tube was sealed and heated at 240 °C for 48 h to achieve sample-spike equilibrium. Osmium was then separated by solvent extraction into CCl_4 (Fujifilm wako infinity pore 99.5+%) and subsequently back-extracted into 47% HBr (TAMAPURE-AA-100). To further purify Os, we employed the microdistillation technique described by ref. 45. The purified Os was loaded onto a platinum filament (Alfa Pt wire), and $Ba(OH)_2$ solution was used as an activator for ionization of Os during measurement. Finally, we analyzed Os trioxide ions by N-TIMS. All data were corrected using procedural blanks for each batch of digestions rather than a long-term average. The average procedural blank for Os was 0.6 ± 0.5 pg ($n = 8$, 1 SD). The blank contribution for measured Os concentrations and ^{187}Os/^{188}Os ratios of KW samples (the low-resolution series) was 0.5–11.2% and 0.3–9.1%, respectively. The blank contribution for measured Os concentrations and

$^{187}Os/^{188}Os$ ratios of KRP and KWR4 samples (the high-resolution series), except KWR4-02 (vein) and KWR4-tuff (tuff), was 0.1–1.2% and 0.03–0.9%, respectively, indicating a negligible effect on the measured values.

We separated Re, Ru, Pd, Ir and Pt from the inverse aqua regia from which Os was removed by the CCl_4 extraction. The separation includes three steps: (1) anion exchange chromatography using AG1-X8 resin; (2) solvent extraction using N-benzoyl-N-phenylhydroxylamine (BPHA) and (3) cation exchange chromatography using AG50W-X8 resin[44]. These elements were analyzed by ICP-QMS. We monitored the masses of ^{89}Y, ^{90}Zr, ^{95}Mo, ^{97}Mo, ^{99}Ru, ^{100}Ru, ^{101}Ru, ^{105}Pd, ^{106}Pd, ^{108}Pd, ^{111}Cd, ^{178}Hf, ^{185}Re, ^{187}Re, ^{191}Ir, ^{193}Ir, ^{194}Pt, ^{195}Pt, ^{196}Pt, and ^{202}Hg, and confirmed that isobaric-oxide interferences on the target elements were negligible. The average blanks for the elements analyzed by ICP-QMS were 0.53 ± 0.16 pg for Ir (n = 7, 1 SD), 0.87 ± 0.46 pg for Ru (n = 7, 1 SD), 8.6 ± 5.2 pg for Pd (n = 7, 1 SD), 15 ± 4 pg for Pt (n = 7, 1 SD), and 3.0 ± 3.2 pg for Re (n = 8, 1 SD). The blank contribution for Re of KW samples was 0.1–6.2%. The blank contribution of KRP and KWR4 samples for Ir, Ru, Pd, Pt, and Re were <11%, <14%, <1.6%, <8.2%, and <0.5%, respectively, except KWR4-02 (vein) and KWR4-tuff.

Due to in-situ decay of ^{187}Re (decay constant of 1.666×10^{-11} year^{-1} [46]) producing ^{187}Os, it was necessary to calculate the initial $^{187}Os/^{188}Os$ ($^{187}Os/^{188}Os_i$) from the measured $^{187}Os/^{188}Os$ value. In this study, the depositional age of all samples was calculated as 66 Ma. If Kawaruppu section were assumed to represent the entire C29R chron and corrected accordingly, the resulting difference in $^{187}Os/^{188}Os$ value compared to assuming a uniform age of 66 Ma would be mostly within 0.03%. Therefore, assuming a uniform age of 66 Ma for all samples has little effect on the results.

Calculation of Os/Ir and $^{187}Os/^{188}Os$ ratios

We performed a simple model calculation of the recovery of the Os/Ir ratio after the K-Pg impact, referring to ref. 10 to estimate the missing interval at the K-Pg boundary. First, the concentration of each element changes over time as follows:

$$\frac{dC}{dt} = \frac{C_{ss} - C}{\tau}$$

where C is the concentration of Os or Ir, t is time, and τ is the marine residence time of Os or Ir. The subscript "ss" denotes steady state. Solving this equation, we obtain:

$$C(t) = \left(C_{ss} - C_0\right) * e^{-t/\tau} \tag{1}$$

here, $t = 0$ is the time of the K-Pg impact, and C_0 is the Os or Ir concentrations of sediment when impact-derived Os and Ir were supplied (i.e., concentration at the K-Pg just boundary). The Os/Ir ratio can be written as follows:

$$\left(\frac{Os}{Ir}\right)_{\text{sediment}} = \frac{\{C_{ss(Os)} - \left(C_{ss(Os)} - C_{0(Os)}\right) * e^{-t/\tau(Os)}\}}{\{C_{ss(Ir)} - \left(C_{ss(Ir)} - C_{0(Ir)}\right) * e^{-t/\tau(Ir)}\}} \tag{2}$$

We applied the data from KRP samples to this equation and simulated it (Fig. 4B), where $C_{ss(Os)}$ and $C_{ss(Ir)}$ were set to 97 pg g^{-1} and 16 pg g^{-1}, respectively, the average of the stable values in Danian. We considered three possible Danian sedimentation rates for the Kawaruppu section:

(1) 30 cm kyr^{-1}, the minimum estimated sedimentation rate (see Supplementary Fig. 6),
(2) 60 cm kyr^{-1}, twice the minimum value and
(3) 90 cm kyr^{-1}, three times the minimum value.

The average Os and Ir concentrations at the K-Pg boundary in hemipelagic sites (Stevns Klint and Caravaca) are 31.9 ng g^{-1} and 46.5 ng g^{-1}, respectively[23]. The sedimentation rate at the K-Pg boundary in the Caravaca section is estimated to be approximately 0.8 cm kyr^{-1} [47]. Using these three assumed sedimentation rates for the Kawaruppu section and these hemipelagic K-Pg data, we estimated the Os and Ir concentrations at the K-Pg boundary in the Kawaruppu section (C_0) under three scenarios. The hemipelagic K-Pg concentrations were diluted according to the relative sedimentation rates, yielding the following C_0 values:

(1) Os = 853 pg g^{-1} and Ir = 1241 pg g^{-1} for a sedimentation rate of 30 cm kyr^{-1};
(2) Os = 426 pg g^{-1} and Ir = 620 pg g^{-1} for 60 cm kyr^{-1}; and
(3) Os = 284 pg g^{-1} and Ir = 413 pg g^{-1} for 90 cm kyr^{-1}.

The oceanic residence times (τ) of Os and Ir were fixed at 40 kyr and 6 kyr, respectively. This choice is based on the results for Site M0077[30], where these residence times provided a good reproduction of the observed Os/Ir ratio variations when C_0 and C_{ss} were fixed to the measured values (see Supplementary Fig. 10).

We also performed a model calculation of the recovery of the $^{187}Os/^{188}Os$ ratio after the K-Pg impact (Fig. 5A). Replacing $C_{ss(Os)}$, $C_{ss(Ir)}$, $C_{0(Os)}$ and $C_{0(Ir)}$ in (2) as $C_{ss(188Os)}$, $C_{ss(188Os)}$, $C_{0(187Os)}$ and $C_{0(188Os)}$, respectively, gives the following formula:

$$R_{(t)} = \left(R_{ss} - \left(R_{ss} - R_0 * E\right) * e^{\left(-\frac{t}{\tau}\right)}\right) / \left(1 - (1 - E) * e^{\left(-\frac{t}{\tau}\right)}\right) \tag{3}$$

where $R_{ss} = C_{ss(^{187}Os)}/C_{ss(^{188}Os)}$, $R_0 = C_{0(^{187}Os)}/C_{0(^{188}Os)}$ and $E = C_{0(^{188}Os)}/C_{ss(^{188}Os)}$.

The seawater Os isotope ratio immediately after the meteorite impact can be expressed as follows:

$$R_0 = f_{ss} * R_{ss} + f_m * R_m$$

where f_{ss}, f_m and R_m are the fraction of Os in seawater derived from the original seawater source after the impact, the fraction of Os in seawater derived from the meteorite after the impact and the $^{187}Os/^{188}Os$ ratio of the meteorite, respectively. Since $f_{ss} + f_m = 1$, the following expression for f_m is obtained:

$$f_m = \left(R_{ss} - R_0\right)/\left(R_{ss} - R_m\right) \tag{4}$$

The seawater ^{188}Os concentration immediately after the impact can be expressed as follows:

$$C_{0(^{188}Os)} = C_{ss(^{188}Os)} + f_m * C_{0(^{188}Os)}$$

Therefore,

$$E = C_{0\left(^{188}Os\right)}/C_{ss\left(^{188}Os\right)} = 1/(1 - f_m) \tag{5}$$

Combining (4) and (5) to eliminate f_m gives

$$E = \left(R_{ss} - R_m\right)/\left(R_o - R_m\right) \tag{6}$$

Here, R_{ss} and R_m were fixed as the average $^{187}Os/^{188}Os_i$ ratio in the Danian of the Kawaruppu section (0.414) and the average value for chondrites (0.127)[48], respectively. The R_o value was set to three patterns: 0.157 (the K-Pg boundary at DSDP 577[8]), 0.200 (the approximate global average of the K-Pg boundary), and 0.251 (the K-Pg boundary at Gubbio[9]). A total of three scenarios were established to simulate the variation in the $^{187}Os/^{188}Os_i$ ratio following the K-Pg impact.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The datasets underlying this research are available as follows: Supplementary Data 1: https://doi.org/10.1594/PANGAEA.994125; Supplementary Data 2: https://doi.org/10.1594/PANGAEA.994126; Supplementary Data 3: https://doi.org/10.1594/PANGAEA.994127; Supplementary Data 4: https://doi.org/10.1594/PANGAEA.994129; Supplementary Data 5: https://doi.org/10.1594/PANGAEA.994130; Supplementary Data 6: https://doi.org/10.1594/PANGAEA.994131.

Received: 21 October 2025; Accepted: 29 April 2026;
Published online: 20 May 2026

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Acknowledgements
We would like to thank Yukari Otsuki, Asuka Tanaka and Junko Kikuchi (JAMSTEC) for supporting Re-PGE analysis. We also thank Drs. Toshihiro Yoshimura and Naohiko Ohkouchi (JAMSTEC) for their support and for providing access to ICP-QMS at JAMSTEC. We gratefully acknowledge the Tokachi General Subprefectural Bureau, Hokkaido Government, for granting us permission for forest entry and rock sample collection. Paleomagnetic measurements were performed under a cooperative research program of the Marine Core Research Institute, Kochi University (22A011, 22B010, 23A002, 23B001, 24A001, 24B001). We thank two anonymous reviewers and Prof. Jaime Urrutia Fucugauchi for their critical but constructive comments on the manuscript.

Author contributions
H.O. and J.K. designed the study and performed PGE and Re–Os isotope analysis. R.T., K. Hayashi, K. Hosogaya, and H.N. performed geological fieldwork and formed the basis of this study. A.I. developed the analytical method of PGEs and K. Suzuki provided laboratory system of Re–Os analyses. R.T., K. Hayashi and B.R.G. performed paleontological analyses. H.H. performed paleomagnetic measurements. Y.O. and M.S. measured U–Pb ages in zircon and provided critical suggestions on geochronological data. H.M., K. Sawada and M.A.I. contributed to the interpretation of geochemical data. All authors contributed to manuscript editing.

Funding
R.T. discloses support for the research of this work from JSPS KAKENHI Grant Numbers JP21H04503, JP18KK0091, JP17K05674 and the Mitsubishi Foundation Grant Number ID27123. J.K. discloses support for the research of this work from JSPS KAKENHI Grant Numbers JP22K18426, JP22H01342, JP23H00153 and JPJSCCA20250004.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43247-026-03602-z.

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Peer review information *Communications Earth and Environment* thanks Mario Fischer, -Godde, Jamie Urrutia-Fucugauchi and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Evan Hastie and Alireza Bahadori. A peer review file is available.

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